

---

# Turn in 4

---

Grant Talbert

February 21, 2025

MA 542 Section A1

Boston University

Note: there is a lot of explicit matrix computation involved in this assignment. I have included all computations at the end just incase you want to see them, but I wasn't sure and didn't want to make it too messy so like yeah.

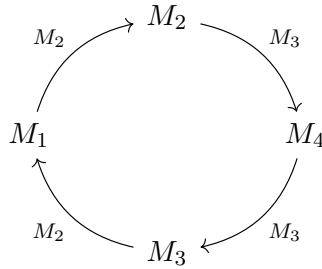
For simplicity, denote

$$M_1 := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad M_2 := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_3 := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad M_4 := \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Also let  $1_R$  denote the identity matrix. Explicit computation yields that

$$M_1 M_2 = M_2, \quad M_3 M_2 = M_4, \quad M_4 M_3 = M_3, \quad M_2 M_3 = M_1.$$

This can be visualized with the diagram:



In some cases the operation is left-multiplication, and in others the operation is right-multiplication. The important thing to note is that if there exists some  $i \in \{1, \dots, 4\}$  with  $M_i \in I$ , then it follows that  $M_i \in I$  for all  $i$ , because  $I$  is an ideal and thus has the absorption property. I will now show the ideal contains at least one of these.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in I$$

be nonzero. Note that  $A = aM_1 + bM_2 + cM_3 + dM_4$ . Since  $A$  is nonzero, at least one element in  $\{a, b, c, d\}$  must be nonzero. If precisely one element is nonzero, then we have some coefficient  $k \in \mathbb{R}$  such that

$$A = kM_i$$

for some  $i$ . Since  $I$  is an ideal and  $\frac{1}{k}1_R \in R$  for  $1_R$  the identity matrix, we have  $\frac{1}{k}1_RA = \frac{k}{k}1_RM_i = M_i$  is an element of the ideal.

Now suppose that more than one coefficient is nonzero. In particular, we must have at least two coefficients are nonzero. Consider the following three cases:

- **Case 1:**  $a, b \neq 0$ , and  $c, d = 0$ . That is,

$$A = aM_1 + bM_2.$$

---

Explicit computation shows that  $M_1M_4 = 0$  and  $M_2M_4 = M_2$ . It follows that

$$AM_4 = (aM_1 + bM_2) = aM_1M_4 + bM_2M_4 = bM_2.$$

Here, I use some facts about vector spaces, such as  $a0 = 0$  for all scalars  $a$ . Since  $I$  is an ideal and  $a \in I$ , we have  $AM_4 = bM_2 \in I$ . By the same logic, we also have

$$\frac{1}{b}1_R(bM_2) = 1_RM_2 = M_2 \in I.$$

- **Case 2:**  $c, d \neq 0$ , and  $a, b = 0$ . That is,

$$A = cM_3 + dM_4.$$

Explicit computation yields that  $M_3M_1 = M_3$  and  $M_4M_1 = 0$ . Therefore, we have

$$AM_1 = (cM_3 + dM_4)M_1 = cM_3M_1 + dM_4M_1 = cM_3.$$

Since  $A \in I$  and  $I$  is an ideal,  $AM_1 = cM_3 \in I$ . Similarly,

$$\frac{1}{c}1_R(cM_3) = M_3 \in I.$$

- **Case 3:** Any other possible scenario. We must have either  $a$ ,  $b$ , or both are nonzero. By explicit computation, we have

$$M_1M_1 = M_1, \quad M_1M_2 = M_2, \quad M_1M_3, M_1M_4 = 0.$$

Since left multiplication fixes  $M_1, M_2$  and annihilates  $M_3, M_4$ , we can take  $c, d = 0$ . If only one of the coefficients  $a, b$  are nonzero, then we have the corresponding  $M_i \in I$  by previous logic. If both are nonzero, apply the same logic as in case 1.

These are the only cases, since  $A \neq 0$ . Therefore,  $M_1, M_2, M_3, M_4 \in I$ .

Since  $I$  is an ideal, it must be a subring, and thus closed under addition. We thus have

$$M_1 + M_4 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 1_R \in I.$$

Let  $M \in \mathcal{M}_2(\mathbb{R})$  be any matrix. Then since  $1_R \in I$ , and  $I$  is an ideal, we have that  $1_RM = M \in I$ . Therefore,  $\mathcal{M}_2(\mathbb{R}) \subseteq I$ . Since  $I$  is an ideal, we have  $I \subseteq \mathcal{M}_2(\mathbb{R})$ , and thus  $I = \mathcal{M}_2(\mathbb{R})$ .